Harbin Institute of Technology (Shenzhen)
Project Report

Image Processing (COMP5033)
2025–2026 Spring Semester
Teacher: Prof. Weizheng Zhang

Project No.:	2	Student Name:	[YOUR NAME]
Student ID:	[YOUR STUDENT ID]	**Date of Submission:**	2026-05-30

---

1 Project Content

This project implements a morphological algorithm to detect the boundaries of all objects in a binary image. The input is *Figure_P2.jpg* (1024×1024 grayscale): a yin-yang design with two interlocking cat silhouettes and a gray background. The pipeline is: (1) binarize the grayscale image using Otsu's automatic thresholding; (2) extract object boundaries via morphological erosion. All key processing is implemented with NumPy; cv2 handles only image I/O.

2 Method Description

2.1 Binarization – Otsu's Method

Otsu's method (1979) finds the threshold T* that maximises the inter-class variance between the two pixel classes (foreground / background):

$$\sigma^2\_B(t) = w\blacksquare(t) \cdot w\blacksquare(t) \cdot [\mu\blacksquare(t) - \mu\blacksquare(t)]^2$$

where $w\blacksquare$, $w\blacksquare$ are class probabilities and $\mu\blacksquare$, $\mu\blacksquare$ are class mean intensities at threshold t. $T^* = \mathrm{argmax}\_t\ \sigma^2\_B(t)$.

Implementation: (1) Compute 256-bin histogram. (2) Scan t from 0 to 255 with running sums for $w\blacksquare$ and cumulative mean. (3) Derive $w\blacksquare = 1-w\blacksquare$ and $\mu\blacksquare$ = (total_mean – cumsum) / $w\blacksquare$. (4) Track maximum σ^2_B. For Figure_P2.jpg: **T* = 107**. Pixels > 107 → foreground (1); pixels ≤ 107 → background (0). Foreground fraction: 60.9%.

2.2 Morphological Boundary Extraction

The **inner boundary** of binary set A with structuring element B is:

$$\beta(A) = A - (A \blacksquare B)$$

where ■ denotes erosion. Erosion shrinks foreground regions by removing boundary pixels; subtracting from the original leaves a 1-pixel-wide boundary.

Erosion (3×3 square SE): (1) Zero-pad binary image by 1. (2) Extract 3×3 sliding windows via *sliding_window_view* (shape H×W×3×3). (3) Output pixel = 1 iff ALL 9 window pixels are 1 (*np.all* over last 2 axes).

To capture boundaries of *both* white and dark regions:

```
boundary_fg = binary – erosion(binary) [white-cat edges] boundary_bg =
(1–binary) – erosion(1–binary) [dark-cat edges] Output = clip(boundary_fg +
boundary_bg, 0, 1)
```

3 Experiment Results and Analysis

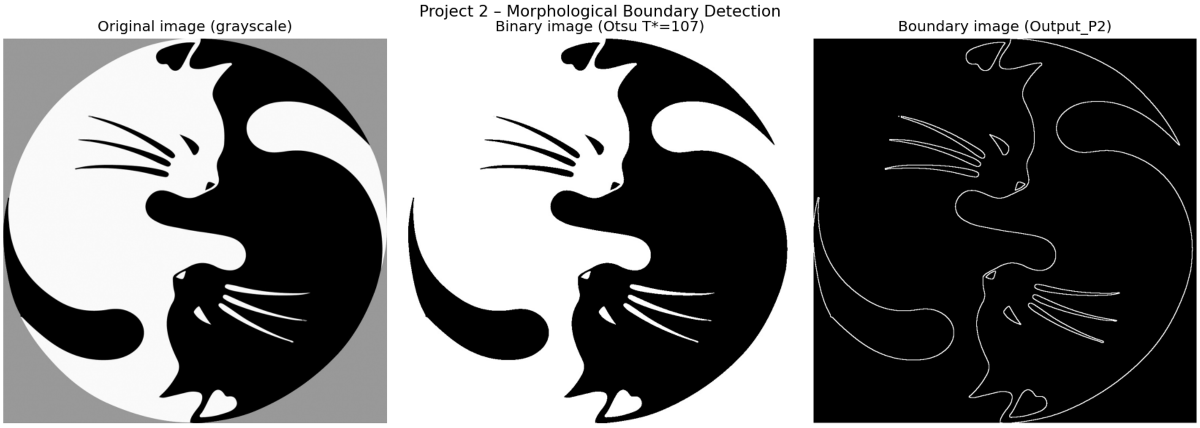

Figure 1. Left: original grayscale image (Figure_P2.jpg). Centre: binary image after Otsu thresholding (T*=107). Right: boundary output (Output_P2.jpg).

Binarization: T*=107 cleanly separates the white cat and fine details from the dark cat body and gray background. Thin whiskers and facial features are well preserved.

Boundary detection: The output contains 27,460 boundary pixels (2.62% of the image). A continuous, 1-pixel-wide boundary accurately traces: the outer circular perimeter, the S-shaped dividing curve, both cat silhouettes, whiskers, ear details, tails, and paw features. No spurious interior edges appear.

Limitations: Very thin structures (whisker lines < 3 px wide) may appear slightly fragmented after 3×3 erosion. A cross-shaped (plus-sign) structuring element would preserve such fine details better.

4 Summary

The main difficulty was correctly binarizing a multi-tone image (gray background + white + black). Otsu's method elegantly solved this by analytically finding T*=107 without manual tuning.

Key knowledge gained: (1) Otsu's thresholding is an efficient unsupervised binarization technique computable in a single O(256) scan of the histogram. (2) Morphological erosion + set subtraction yields precise 1-pixel boundaries without complex edge operators. (3) Processing both the binary image and its complement captures all object edges in a single boundary map. (4) Structuring element choice (size/shape) controls boundary thickness and sensitivity to fine structures.